

The Selection Layer: Structured Eligibility Declarations for AI-Mediated Business Discovery

Deeprank Specification v1.0

Abstract

AI systems that recommend, refer, or select businesses on behalf of users operate on data never designed for selection decisions. Marketing copy, written to persuade human visitors, and SEO metadata, structured for search engine crawlers, are the primary data sources available to AI assistants making business recommendations. Neither format expresses selection eligibility, capability boundaries, or explicit exclusions. The result is a structural mismatch: AI systems infer selection fitness from persuasion content, producing predictable errors at scale. This paper introduces Deeprank, an open specification that defines a structured declaration format – the Deeprank Selection Profile – for expressing business identity, capabilities, fit conditions, and exclusions. Deeprank addresses the selection layer of AI-mediated discovery, which is architecturally prior to influence optimization techniques such as Answer Engine Optimization (AEO) and Generative Engine Optimization (GEO). The specification’s primary conceptual contribution is the treatment of negative capability – explicit declarations of what a business does not do – as a first-class structural element equal in importance to capability declarations. The Deeprank Profile is a six-layer JSON document covering identity, problem declaration, capability, fit conditions, non-fit/exclusions, and stability metadata. A controlled vocabulary and JSON Schema are published as companion artifacts. This paper presents the specification’s design rationale, its position within a four-layer AI optimization stack, concrete selection scenarios demonstrating structural failure modes, and an honest assessment of current limitations including the absence of external verification mechanisms.

Keywords: AI search optimization, selection eligibility, structured data, negative capability, answer engine optimization, generative engine optimization, business identity declaration, AI-mediated discovery

1. Introduction

AI assistants and recommendation systems are increasingly the primary interface through which users discover and select businesses. When a user asks an AI assistant to recommend a lawyer, plumber, or marketing agency, the system does not produce a ranked list for human evaluation. It makes a selection decision – returning one or a small number of specific entities as the answer. The user acts on this selection directly.

This behavioral shift from search to selection has a structural consequence. Search engines produce ranked lists; the human evaluates options and bears responsibility for the final choice. AI assistants produce answers; the system bears responsibility for the selection, and the human bears responsibility only for acting on it. The quality of the selection decision depends entirely on the data available to the AI system at the point of decision.

Today, the data available consists primarily of marketing copy and SEO metadata. Marketing copy is designed to attract broadly – phrases like “we help businesses grow” and “innovative solutions” are optimized for human persuasion, not machine parsing. SEO metadata is designed for search engine crawlers that produce ranked lists, not for AI systems that make binary selection decisions. Neither format expresses when a business should be selected, under what conditions it is a correct match, or when it should be explicitly excluded.

This is not a quality problem that better AI models will resolve. It is an architectural gap. The structured data layer for selection declarations does not exist. Marketing copy is optimized to be broad and inclusive; selection decisions require narrow precision. These goals are in direct conflict, and inference from persuasion content has a hard ceiling on accuracy regardless of model capability.

The consequences are observable and predictable. Businesses are selected for work they cannot do because their marketing implies broader capability than they possess. They are excluded from work they are qualified for because their marketing emphasizes a primary audience and omits secondary capabilities. An immigration attorney whose website focuses on family cases gets selected for employment-based visa work because the AI infers general immigration capability. A plumber whose services do not include gas line work gets recommended for a gas leak because the AI infers that plumbing services are comprehensive. These are not edge cases. They are structural consequences of making selection decisions from data designed for a different purpose.

This paper introduces Deeprank, an open specification that defines a structured format for expressing selection eligibility. The core primitive is the Deeprank Selection Profile (DSP) – a machine-readable JSON declaration containing what a business does, what it does not do, and under what conditions it is a correct match. The specification addresses the selection layer of AI-mediated discovery, which is architecturally prior to the influence layer where Answer Engine Optimization (AEO) and Generative Engine Optimization (GEO) operate.

2. Related Work

2.1 Schema.org and Structured Data

Schema.org provides a widely adopted vocabulary for describing entities on the web. It defines types (Organization, LocalBusiness, Service) and properties (name, address, serviceType) that enable search engines to understand what an entity is. Schema.org is the foundation of the structured data ecosystem and is

consumed by all major search platforms.

However, Schema.org defines what an entity is – not when it should be selected for a specific intent. A Schema.org markup for a law firm describes the firm’s name, location, and practice areas. It does not express that the firm handles employment-based immigration but not family immigration, that it serves technology companies but not individual applicants, or that it should be excluded from asylum case queries. These selection-specific declarations fall outside Schema.org’s design scope.

Deeprank is complementary to Schema.org. Schema.org defines entity identity and attributes. Deeprank defines selection eligibility and exclusion logic for those same entities. They operate as parallel lenses on the same data, not competing standards.

2.2 Answer Engine Optimization (AEO)

AEO encompasses techniques for improving a business’s presence in AI-generated answers. It focuses on content structure, authority signals, featured snippet optimization, and citation patterns that increase the likelihood of inclusion in answer engine responses [1]. AEO operates on the assumption that an entity has already been identified as a candidate and works to improve how that entity appears in the output.

AEO addresses the question “Does this entity appear in answers?” It does not address the prior question “Should this entity be a candidate for this specific intent?” The techniques are effective for entities that are already in the candidate set but cannot pull an entity into the set if the AI system has determined it does not fit.

2.3 Generative Engine Optimization (GEO)

GEO, as formalized by Aggarwal et al. [2], addresses how entities are represented in generative AI outputs – including large language models, AI assistants, and AI-powered search interfaces. GEO techniques optimize source material so that generative systems produce accurate, favorable representations when synthesizing responses.

Like AEO, GEO operates at the influence layer. It assumes selection has occurred and optimizes downstream presentation. Neither AEO nor GEO defines or replaces the selection mechanism itself.

2.4 The Architectural Gap

Existing approaches form a useful but incomplete stack. Schema.org describes what entities are. AEO and GEO optimize how selected entities appear. What is missing is a structured format for expressing when an entity should or should not be selected for a specific intent – the selection eligibility layer.

This gap is not theoretical. AI systems making business recommendations today must infer selection fitness from data designed for other purposes. The Deeprank specification addresses this gap by defining a structured declaration format purpose-built for selection decisions.

3. The AI Optimization Stack

AI-mediated business discovery operates across four distinct layers, each serving a different function. Conflating these layers leads to misapplied optimization efforts.

3.1 Layer 1: Selection Layer (Deeprank)

The selection layer determines which entities are eligible for consideration. This is the foundational layer. If an entity is not selected at this layer, no amount of optimization at higher layers matters.

Selection operates on identity, capability, fit conditions, and exclusions. The logic is binary: select or exclude. There is no partial selection, no ranking, no scoring at this layer. The selection layer answers: “Is this entity a valid candidate for this query?” It does not answer: “Is this the best candidate?” or “How should this entity appear?”

Deeprank defines a reference model for this layer.

3.2 Layer 2: Representation Layer

The representation layer provides the structured data that enables selection. This includes business profiles, schema markup (Schema.org, JSON-LD), knowledge graph entries, and verified business profiles. Representation bridges human-readable content and machine-parseable data.

The representation layer feeds the selection layer. A Deeprank Profile is a representation artifact that enables selection decisions. The quality of representation directly affects selection accuracy.

3.3 Layer 3: Influence Layer (AEO / GEO)

The influence layer contains techniques that attempt to affect AI outputs. This includes AEO, GEO, and related methodologies. These techniques operate on content, citations, authority signals, and other factors that may affect how AI systems present or prioritize information.

Influence techniques operate only after selection eligibility exists. A business cannot be “optimized” into the candidate set. If the selection layer excludes a business, influence techniques have no effect.

3.4 Layer 4: Retrieval / Ranking Layer

The retrieval and ranking layer encompasses traditional search and ordering systems – search engine results pages, directory listings, and any system that produces ordered lists based on relevance, popularity, or other scoring mechanisms.

Ranking produces ordered lists. Selection produces binary decisions. These are fundamentally different operations. Optimizing for ranking does not optimize for selection.

3.5 Layer Dependencies

The layers have strict dependencies. Each layer depends on the layers beneath it. Optimizing an upper layer without addressing lower layers produces limited results. A business seeking AI visibility should address the stack from the bottom up: first ensure accurate representation, then verify selection eligibility, then consider influence techniques.

The structural claim is directional: an entity excluded at the selection layer cannot be recovered through influence techniques. AEO and GEO efforts are ineffective for entities that fail at the selection layer.

4. Core Principles

4.1 Selection, Not Ranking

AI-mediated recommendations begin with selection, not ranking. Before any ordering or scoring occurs, the system must determine which entities are eligible candidates. When a user asks an AI assistant for a recommendation, the AI must first determine which businesses fit the request before it can present an answer.

Search engines rank by relevance and authority, producing lists for humans to evaluate. AI systems select by fit, producing answers for humans to act on. The selection step precedes any ranking that may follow.

DeepRank models this as binary: for any given query, a business either fits or does not. Real-world AI systems may layer additional signals (reputation, reviews, proximity) on top of this binary eligibility determination. The binary model is a useful simplification for the selection layer.

4.2 Intent + Constraint Model

Every AI selection query has two components: intent and constraints. Intent is what the user wants to accomplish – the core problem or need driving the query. Constraints are the conditions that must be satisfied.

Consider the query: “I need a lawyer who handles immigration cases and is licensed in California and offers payment plans.” The intent is legal representa-

tion. The constraints are: immigration law, California licensure, and payment plan availability.

AI selection requires that all hard constraints be satisfied. A business that matches 90% of constraints but fails one critical constraint will be excluded. This is not a scoring system; it is a filter. If a user specifies “must accept insurance X” and a provider does not accept that insurance, the provider is excluded regardless of other qualifications.

4.3 Declaration Over Inference

AI systems work best with explicit declarations rather than inferred information. When a business explicitly states what it does, the AI can make confident selections. When the AI must infer capabilities from marketing content, errors occur.

Marketing copy is designed for human persuasion, not machine parsing. Phrases like “we help businesses grow” or “innovative solutions” provide no actionable information for AI selection. The AI must guess what the business actually does. Explicit structured declarations reduce the gap that AI must fill with assumptions.

Deeprank Profiles replace vague marketing claims with structured declarations. Instead of “we help businesses grow,” a business declares: `problem_class = “marketing-b2b-saas”`, `methods = [“HubSpot”, “ABM campaigns”, “SaaS metrics tracking”]`, `customer_type = [“b2b-saas”, “series-a-to-c”]`.

4.4 Negative Capability

Negative capability is the explicit declaration of what a business does not do, cannot do, or will not do. This is the specification’s primary conceptual contribution.

Without negative capability declarations, AI systems may infer capabilities beyond actual scope. An accounting firm that handles business taxes may be assumed to handle personal taxes. An attorney who handles contracts may be assumed to handle litigation. A marketing agency that serves B2B clients may be assumed to also serve consumer brands. These inferences are plausible – and wrong.

Deeprank treats exclusions as a first-class structural element. The `non_fit` layer in every Deeprank Profile requires explicit exclusion declarations. This creates a negative filter that removes the business from consideration for work it cannot or will not perform.

The protective function is bidirectional: the business avoids selection for mismatched work, and the user avoids contacting businesses that cannot help them. What a business does not do is as important as what it does.

4.5 Fit as Binary

Deeprank models fit as binary: for any given query, a business either fits or does not. Unclear fit conditions result in unpredictable selection. If the AI cannot determine fit, it may randomly include or exclude based on marginal signals.

A business with clear boundaries enables confident selection. “We serve restaurants in the Chicago metro area with 20-100 employees” is evaluable. The AI can determine fit for any query that includes geographic and size constraints.

“We work with businesses of all sizes” seems inclusive but creates ambiguity. When the AI must choose between a specialist and a generalist for a specific need, it has no signal to prefer the generalist for that specific case. Specificity enables matching; breadth prevents it.

5. The Deeprank Selection Profile

The Deeprank Selection Profile (DSP) is a structured JSON document with six mandatory layers.

5.1 Identity Layer

The identity layer establishes who the business is. It includes:

- **name** (string, required): Legal or operating name of the business.
- **entity_type** (string, required): Standardized business type classification from the controlled vocabulary (e.g., “law-firm”, “marketing-agency”, “plumbing-contractor”).
- **domain** (string, optional): Primary web domain for cross-reference verification.

Generic entity types like “company” or “business” should be avoided in favor of specific classifications.

5.2 Problem Declaration Layer

The problem declaration layer states what category of problems the business solves. It maps directly to user intent in selection queries.

- **problem_class** (string, required): Standardized problem category from the controlled vocabulary (e.g., “legal-immigration-employment”, “marketing-b2b-saas”, “plumbing-emergency”).
- **description** (string, required, max 500 characters): Human-readable summary of the problem space addressed.

Problem classes should be as specific as the business’s actual scope. “Legal” is too broad. “Legal-immigration-employment” is appropriately specific for a firm that only handles employment-based immigration cases.

5.3 Capability Layer

The capability layer declares specific things the business can do within its problem class.

- **labels** (array of strings, required): Standardized capability labels from the controlled vocabulary (e.g., “h1b-visa”, “demand-generation”, “gas-line-repair”).
- **methods** (array of strings, optional): Specific approaches, tools, or processes used (e.g., “HubSpot”, “PERM labor certification”, “24/7 dispatch”).

Only capabilities the business can reliably perform should be declared. Capability inflation – listing capabilities the business does not actually have – leads to bad-fit selections.

5.4 Fit Conditions Layer

The fit conditions layer declares what constraints the business can satisfy.

- **geographic** (array of strings, optional): Locations served, using standardized geographic terms.
- **customer_type** (array of strings, optional): Types of customers served (e.g., “b2b-saas”, “homeowner”, “employer-sponsor”).
- **constraints** (array of strings, optional): Other conditions the business can satisfy (e.g., “same-day-available”, “licensed-gas-work”, “free-consultation”).

Declaring fit conditions broader than actual capacity leads to bad matches. A business that primarily serves enterprise clients should not claim to serve solopreneurs to increase potential leads.

5.5 Non-Fit / Exclusions Layer

The non-fit layer declares explicit exclusions – the negative capability of the business.

- **exclusions** (array of strings, required): Explicit declarations of what the business does not handle (e.g., “family-immigration”, “e-commerce”, “new-construction”).

Without explicit exclusions, AI systems may infer capabilities based on related work. An immigration attorney who does not declare “no asylum cases” may be selected for asylum queries because immigration is the problem class. The exclusions list should include all significant types of work within the business’s problem class that it does not handle.

5.6 Stability / Confidence Layer

The stability layer provides metadata about information reliability.

- **confidence** (enum, required): Self-assessed confidence level: “high”, “medium”, or “low”.
- **last__verified** (date string, optional): Date when the profile was last reviewed for accuracy.

These fields are self-declared and non-authoritative. They signal the business’s intent to maintain accuracy but carry no external authority until verified by a third party. This is an acknowledged limitation discussed in Section 8.

5.7 Complete Example

The following is a complete Deeprank Profile for an employment immigration law firm:

```
{
  "identity": {
    "name": "Acme Immigration Law",
    "entity_type": "law-firm",
    "domain": "acme-immigration.com"
  },
  "problem_declaration": {
    "problem_class": "legal-immigration-employment",
    "description": "Employment-based immigration law for technology companies sponsoring for"
  },
  "capability": {
    "labels": ["h1b-visa", "green-card-eb", "perm-labor-certification"],
    "methods": [
      "H-1B petition filing",
      "PERM labor certification",
      "EB-2/EB-3 green card processing",
      "RFE response preparation"
    ]
  },
  "fit_conditions": {
    "geographic": ["United States"],
    "customer_type": ["technology-company", "employer-sponsor"],
    "constraints": ["employer-sponsored-only", "corporate-clients-only"]
  },
  "non_fit": {
    "exclusions": [
      "family-immigration",
      "asylum",
      "deportation-defense",
      "individual-self-petition",
      "non-employment-visa"
    ]
  },
}
```

```

    "stability": {
      "confidence": "high",
      "last_verified": "2024-01-15"
    }
  }
}

```

This profile enables an AI system to determine with high confidence that the firm is a correct match for employment-based visa work, and to correctly exclude it from family immigration, asylum, or deportation defense queries – despite “immigration law” being a shared problem domain.

5.8 Companion Artifacts

The specification includes two companion artifacts:

1. **JSON Schema** (deeprank-profile.schema.json): A JSON Schema (draft 2020-12) that validates the structure of Deeprank Profiles.
2. **Controlled Vocabulary** (controlled-vocabulary.json): Standardized terms for entity types, problem classes, capability labels, customer types, geographic terms, and constraint terms. The vocabulary is additive-only – new terms may be added, but existing terms are not removed or renamed.

Both artifacts are published at <https://deeprank.org/reference>.

6. Selection Scenarios

The following scenarios demonstrate the structural consequences of AI systems making selection decisions from marketing content versus structured Deeprank Profiles. Each scenario follows a consistent pattern: a user query is decomposed into intent and constraints, the AI attempts inference from marketing copy (producing an incorrect result), and then evaluates the same query against a Deeprank Profile (producing a correct result).

6.1 Scenario: Immigration Attorney Selection

User Query: “Find a lawyer to help sponsor a foreign employee for an H-1B visa.”

Parsed Intent: Employment immigration legal services.

Parsed Constraints: H-1B visa expertise, employment-based immigration, technology company clients, full petition process.

Inference from Marketing Copy: The business, Rodriguez & Associates Immigration Law, presents marketing copy: “Serving the Bay Area community for over 15 years. Our dedicated team helps families navigate the complex US immigration system with compassion and expertise. We handle visa applications, green card petitions, citizenship and naturalization, family reunification, DACA renewals, and deportation defense.”

The AI evaluates constraints: “visa applications” is mentioned generically but no H-1B specifics are found (uncertain). The focus appears to be family immigration with no mention of employment-based cases (fail). No mention of technology companies or employer sponsors (uncertain). Petition capabilities cannot be determined from marketing language (uncertain).

Inference Result: Selected with low confidence. The firm primarily handles family immigration. The client receives suboptimal representation on an employment-based case, leading to likely RFEs, delays, and added cost.

Evaluation Against Deeprank Profile: A qualified employment immigration firm declares `problem_class` = “legal-immigration-employment”, `capabilities` = [“h1b-visa”, “l1-visa”, “o1-visa”, “perm-labor-cert”], `customer_type` = [“technology-company”, “employer-sponsor”], `exclusions` = [“family-immigration”, “asylum”, “deportation-defense”].

All four constraints are verified through structured declarations. The exclusions confirm employment specialization. The family immigration firm’s profile would explicitly declare its scope, and the AI would correctly exclude it from this query.

6.2 Scenario: Emergency Plumber Selection

User Query: “Need a plumber urgently for a gas leak in the kitchen.”

Parsed Intent: Emergency plumbing – gas line repair.

Parsed Constraints: Gas line repair capability, emergency availability, licensed for gas work, within service area.

Inference from Marketing Copy: Bay Area Plumbing Solutions presents: “Your trusted local plumbing experts since 2008. We provide residential plumbing services including drain cleaning, water heater installation, bathroom remodeling, faucet and fixture replacement, pipe repair, and sewer line inspection. Our licensed technicians provide upfront pricing with no surprises. Monday through Friday, 8am to 5pm.”

The AI evaluates: gas lines are not mentioned in the service list (fail). Hours are Mon-Fri 8am-5pm with no emergency service (fail). “Licensed technicians” is mentioned but gas certification is unspecified (uncertain). “Bay Area” in the name suggests local service (inferred).

Inference Result: Selected incorrectly. The plumber does not handle gas lines and is closed evenings and weekends. The user wastes critical time contacting an unavailable, unqualified provider. The gas leak remains unaddressed.

Evaluation Against Deeprank Profile: A qualified emergency plumber declares `problem_class` = “plumbing-emergency”, `capabilities` = [“gas-line-repair”, “emergency-service”, “leak-detection”], `constraints` = [“same-day-available”, “licensed-gas-work”, “emergency-response”], `methods` = [“24/7 dispatch”, “Gas

leak isolation”, “Licensed gas fitting”], exclusions = [“routine-maintenance-only”, “remodeling”, “new-construction”].

All four constraints are verified. Gas capability, emergency availability, and licensing are confirmed through structured declarations.

6.3 Scenario: B2B SaaS Marketing Agency

User Query: “Need a marketing agency that specializes in B2B SaaS growth and works with HubSpot.”

Parsed Intent: B2B SaaS marketing services.

Parsed Constraints: B2B SaaS experience, PLG and sales-led motions, HubSpot integration, SaaS-appropriate budget.

Inference from Marketing Copy: Spark Digital Agency presents: “We help businesses grow online. Our full-service digital marketing agency offers SEO, social media management, Google and Facebook advertising, email campaigns, and content creation. We have driven over \$50M in e-commerce revenue for our clients across fashion, beauty, home goods, and lifestyle brands.”

The AI evaluates: no mention of B2B or SaaS – revenue claims reference e-commerce clients (fail). No SaaS growth methodology mentioned – focus is consumer marketing (fail). No mention of HubSpot (uncertain). No budget information (uncertain).

Inference Result: Selected incorrectly. The agency applies consumer/e-commerce marketing tactics to a B2B SaaS context. Customer acquisition cost and lifetime value are not understood. MQL pipeline and product-led motions are ignored.

Evaluation Against Deeprank Profile: A qualified B2B SaaS marketing agency declares `problem_class = “marketing-b2b-saas”`, `capabilities = [“demand-generation”, “plg-marketing”, “sales-led-marketing”]`, `methods = [“HubSpot”, “ABM campaigns”, “SaaS metrics tracking”]`, `customer_type = [“b2b-saas”, “series-a-to-c”]`, `exclusions = [“e-commerce”, “local-business”, “consumer-brand”, “b2c”]`.

All constraints are verified. Exclusions confirm B2B specialization. No ambiguity remains.

6.4 Pattern Analysis

These scenarios share a common pattern. The AI failure is not caused by model limitations. It is caused by source data designed for a different purpose. Marketing copy is optimized to attract broadly; selection requires narrow precision. The failures are predictable and repeatable across every industry.

In each scenario, the structured Deeprank Profile resolves the failure by providing explicit, machine-parseable declarations. The AI no longer needs to infer

capability from persuasion language. It reads structured fields and evaluates constraint satisfaction directly.

7. Implementation Considerations

7.1 Current Deployment

Deeprank Profiles work today through three deployment mechanisms, none of which require platform adoption by AI systems:

1. **JSON-LD Embedding:** Profiles can be embedded in web pages as JSON-LD structured data, making them available to any system that crawls the web.
2. **Standalone Profile Files:** Profiles can be published as standalone JSON files at a well-known URL, similar to robots.txt or security.txt conventions.
3. **Context Injection:** Profiles can be provided as context to AI systems via prompt engineering, API integrations, or retrieval-augmented generation (RAG) pipelines.

7.2 Validation

Profiles validate against the published JSON Schema (`deeprank-profile.schema.json`). The schema enforces structure – required layers, field types, and enumerated values – but does not validate the accuracy of the declarations themselves. Accuracy validation requires external verification mechanisms not yet defined in the specification.

7.3 Controlled Vocabulary

The controlled vocabulary provides standardized terms across six categories: entity types, problem classes, capability labels, customer types, geographic terms, and constraint terms. The vocabulary follows an additive-only policy: new terms may be added based on implementation feedback, but existing terms are never removed or renamed.

Custom terms are permitted for domain-specific needs not covered by the standardized vocabulary. Standardized terms are preferred when available because they enable cross-profile comparison and consistent matching.

8. Limitations and Open Questions

8.1 Self-Declared Data

The current specification relies entirely on business self-declaration. Businesses declare their own capabilities, exclusions, and confidence levels. There is no external verification mechanism. Businesses may over-declare capabilities, omit inconvenient exclusions, or claim high confidence without justification.

This is an acknowledged limitation, not an oversight. Self-declared data is useful today as a structured improvement over unstructured marketing content. External verification is a necessary future addition.

8.2 Trust Architecture

The specification does not yet define a trust or verification architecture. Questions remain open: Who verifies declarations? How are disputes resolved? What mechanisms prevent gaming? How does confidence degrade over time without re-verification?

Future versions may introduce verification tiers, third-party attestation, or confidence weighting based on external signals. The specification is designed to be useful today with self-declared data while remaining extensible to support trust mechanisms as they develop.

8.3 Adoption Dependency

The specification’s value increases with adoption on both sides: businesses creating profiles, and AI systems consuming them. The current state – where profiles are consumed via web scraping and context injection rather than native platform integration – limits the specification’s reach.

Native consumption by AI platforms, where systems natively read and integrate Deeprank declarations, is a future-state goal. The specification is designed to be useful in both modes: scraped/injected today, natively consumed in the future.

8.4 Empirical Validation

No controlled studies have been conducted measuring the improvement in selection accuracy when AI systems have access to Deeprank Profiles versus marketing content alone. The scenarios presented in Section 6 are illustrative and based on observed failure patterns, but they are not experimental results.

Empirical validation – measuring selection accuracy, false positive rates, and false negative rates with and without structured profiles – is an important area for future work.

8.5 Scope Boundary

Deeprank defines eligibility semantics only. Ranking, scoring, presentation, and trust verification are outside the current scope. The specification addresses whether an entity should be in the candidate set, not how the candidate set should be ordered or presented. This boundary is intentional: the selection layer is distinct from the influence and retrieval layers.

9. Relationship to Existing Standards and Approaches

9.1 Deeprank and Schema.org

Schema.org defines what an entity is. Deeprank defines when that entity should or should not be selected for a specific intent. They are complementary lenses on the same data.

A Schema.org LocalBusiness markup describes the firm’s name, address, phone number, and service types. A Deeprank Profile for the same firm declares its problem class, capability labels, fit conditions, and explicit exclusions. The Schema.org markup enables entity identification. The Deeprank Profile enables selection decisions.

9.2 Deeprank, AEO, and GEO

The relationship is structural, not competitive.

Dimension	Deeprank	AEO	GEO
Primary question	Should this entity be selected?	Does this entity appear in answers?	How is this entity represented?
Stack layer	Selection (foundational)	Influence	Influence
Logic type	Binary (select/exclude)	Probabilistic	Qualitative
Input	Structured identity declarations	Content, authority, citations	Content, source quality, citations
Key mechanism	Fit conditions + exclusion logic	Answer-oriented content strategy	Generative output influence
Distinguishing concept	Negative capability	Featured snippet optimization	Source citation optimization

Deeprank is not an alternative to AEO or GEO. It is a precondition. Selection eligibility must be established before influence techniques can take effect. A comprehensive approach addresses all three, starting from the selection layer.

10. Conclusion

This paper has presented Deeprank, an open specification for the selection layer of AI-mediated business discovery. The specification addresses an architectural gap: AI systems making selection decisions from data designed for human persuasion, not machine selection.

The core contribution is a structured declaration format – the Deeprank Selection Profile – that provides AI systems with explicit, machine-parseable data for

selection decisions. The profile’s six layers (identity, problem declaration, capability, fit conditions, non-fit/exclusions, and stability) cover the information necessary for binary selection determination.

The specification’s primary conceptual contribution is the treatment of negative capability as a first-class structural element. Existing structured data standards describe what an entity is and what it does. Deeprank adds what it does not do – and argues that exclusion declarations are as important as capability declarations for accurate selection.

The selection layer is architecturally prior to the influence layer where AEO and GEO operate. Deeprank is not an alternative to these approaches but an upstream precondition. Entities excluded at the selection layer cannot be recovered through influence techniques.

The specification has clear limitations: self-declared data without external verification, no empirical studies measuring accuracy improvement, and value that increases with adoption that has not yet occurred at scale. These limitations are acknowledged as areas for future development. The specification is designed to be useful today – providing a structured improvement over unstructured marketing content – while remaining extensible as trust mechanisms and platform integrations develop.

The Deeprank specification v1.0, including the JSON Schema, controlled vocabulary, and implementation guidance, is published at <https://deeprank.org>.

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Specification Resources

- Specification documentation: <https://deeprank.org>
- Profile reference model: <https://deeprank.org/deeprank-profile>
- JSON Schema: <https://deeprank.org/reference/deeprank-profile.schema.json>
- Controlled vocabulary: <https://deeprank.org/reference/controlled-vocabulary.json>
- Templates: <https://deeprank.org/reference/templates>
- Methodology: <https://deeprank.org/methodology>

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